ECS Transactions

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To cite this article: Charles R. Sullivan *et al* 2011 *ECS Trans.* **41** 315

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ECS Transactions, 41 (8) 315-330 (2011)
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Passive Component Technologies for Advanced Power Conversion Enabled by Wide-Band-Gap Power Devices

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Modern power electronics typically operate with switching frequencies in the tens of kilohertz to a few megahertz. SiC and GaN power switches promise to enable increasing these frequencies by an order of magnitude or more. Doing so would allow reducing the energy storage requirements for passive components, particularly transformers and inductors, which typically take up the largest portion of the volume of a power converter, and which are also responsible for much of the power loss, cost, and design difficulty in the converter. In principle, a smaller energy storage requirement should make these components smaller and lower in loss and cost. However, material, fabrication, and design constraints make it hard to achieve such improvements. These issues are reviewed and new results on design optimization of low profile air-core inductors and loss characterization of high-performance magnetic materials for 10 to 100 MHz are presented.

Introduction

Modern power electronics typically operate with switching frequencies in the tens of kilohertz to a few megahertz. SiC and GaN power switches promise to enable increasing these frequencies by an order of magnitude or more. Doing so would allow reducing the energy storage requirements for passive components—inductors and capacitors—which typically take up the largest portion of the volume of a power converter, and which are also responsible for much of the power loss, cost, and design difficulty in the converter. In principle, a smaller energy storage requirement should make these components proportionally smaller, and likewise reduce their cost and power loss. However, materials, fabrication methods, and design approaches that work well for large components in the kilohertz range face challenges in the megahertz range, potentially preventing the realization of the benefit that should be provided by higher switching frequencies. In this paper, we review technologies for passive components for efficient power conversion in the 3-300 MHz frequency range focusing on inductors and transformers. As they do at lower frequencies, these magnetic components pose more significant challenges than capacitors do, and we present new results on several approaches to magnetic components.

Between 100 kHz and 1 MHz, MnZn ferrites are the preferred magnetic material for transformers and inductors because they offer low power losses at moderate flux densities in this frequency range. However, their losses increase rapidly above 1 MHz. Attractive alternatives for higher frequencies include NiZn ferrites and thin-film sputtered nano-granular materials. We examine performance of these materials, including new measurements on the nickel zinc ferrite that emerged as the most attractive option based on the characterization work in [1,2].

Thin-film sputtered nano-granular materials, such as Co-Zr-O, can have lower losses and support higher flux densities than ferrites. Permeability is maintained to several gigahertz, and low-loss operation is possible up to at least 100 MHz. These performance attributes allow miniaturizing magnetic components, and the deposition process for these materials is compatible with microfabrication processes, making it possible to integrate magnetics with a semiconductor integrated circuit on or over die[3,4,5,6]. To improve power handling and efficiency of such components, we report scaling up deposition of Co-Zr-O nano-granular magnetic materials from a previous maximum thickness of 10 μm to 19.3 μm, while maintaining good soft magnetic properties. We also report new measurements on the transformers developed in[6] with high-power excitation confirming that the material is able to operate at high drive levels without significant performance degradation.

Despite the advantages of magnetic materials, and progress in their performance, all magnetic materials have performance that deteriorates as frequency increases. In contrast, the performance of air-core magnetics improves monotonically with frequency[7,8]. Thus, for sufficiently high frequencies, air-core magnetics can become attractive. We have previously reported optimization of toroidal air-core inductors, and a simple calculation of the maximum achievable quality factor that applies to both toroidal and solenoid inductors, given a thickness constraint and no constraint on footprint area. Here, we introduce a general calculation for both types, considering a constraint on footprint area.

High-Frequency Windings

Any magnetic component, whether it is an inductor or a transformer, must have one or more windings that link a flux path, whether that flux path is through a soft magnetic material or through air or other non-magnetic material. Increasing the frequency results in a decrease in the electromagnetic skin depth, given by $\delta = \sqrt{\rho/(\pi f \mu_0)}$ where ρ is the resistivity, of the conductor, f is the frequency of the current waveform, and μ_0 is the permeability of the conductor, equal to the permeability of free space in most conductors. For copper with a resistivity of 1.72 $\mu\Omega$cm, the skin depth is 21 μm at 10 MHz, and 6.6 μm at 100 MHz. If a conductor is thick compared to these dimensions, current will only flow on the surface. Although the actual behavior is an exponential decay of current density into the conductor, for many purposes a conceptual model of current flowing in a layer one skin depth thick not only provides correct conceptual understanding, but also provides an accurate power loss calculation.

However, an understanding of skin effect alone is not adequate to design a low-loss winding at high frequency. A conductor in an inductor or transformer is subjected to fields produced by other conductors in the component, which can induce complex eddy currents. This *proximity effect* is often responsible for a bigger increase in losses and effective ac resistance than skin effect[9,10]. In general, there are two approaches to reducing proximity effect. One is to use conductors that are thin compared to a skin depth, such that there is insufficient room for eddy currents to circulate within a conductor. In wire-wound components, a typical approach to realize this is to use litz wire—wire made up of many fine strands, individually insulated[11]. It can also be done with thin foil layers of conductor material[12,13,14]. In either case, the goal is to end up with dimensions that are thin compared to a skin depth in the direction perpendicular to the magnetic field direction, and to stack multiple layers in this direction, with current forced to share equally between these layers. A second approach is to configure the winding in a single layer, such that current can flow on the surface of that layer, in a region one skin-depth deep. The resulting resistance is

worse than would be predicted based only on skin effect, because the current only flows on one side of a given wire, rather than on its whole perimeter, but the worst proximity-effect losses, which can increase resistance by several orders of magnitude, are avoided.

A general comparison of these two approaches to reducing proximity-effect losses – multi-layer and single-layer – shows that, in the simplest cases, a multi-layer winding with conductor thicknesses optimized for one frequency can reduce the ac resistance relative to a single-layer winding by a factor of $1/\sqrt{p}$, where p is the number of layers in the winding, if the correct layer thickness, which can be approximated by

$$t \approx \frac{1.3\delta}{\sqrt{p}}, \tag{1}$$

is used[15,16,17]. Achieving this requires ensuring that the current is shared equally between the layers. This is achieved by twisting in litz wire, but requires other strategies in planar conductors[12,13,14].

At frequencies in the megahertz range, using litz wire becomes challenging, because of the conductor dimensions required. Although magnet wire as small as 12 μm or AWG 56 is available, it is prohibitively expensive and difficult to work with in diameters smaller than about 30 μm (about AWG 48). If the number of winding layers is chosen to minimize loss for a given wire diameter (rather than vice versa), the amount that the loss is reduced compared to using a thick single-layer winding can be expressed as[15]

$$\frac{P_{ml}}{P_{sl}} \approx 0.584 \left(\frac{d_{min}}{\delta} \right). \tag{2}$$

For example, for 10 MHz, 20 μm (AWG 52) has a diameter about equal to the skin depth, and could be used to achieve a 39% reduction in winding loss relative to a single-layer design.

Given the limited advantages of practical litz-wire at 10 MHz or above, it is interesting to consider using conductor materials other than copper: a higher-resistivity material would have a thicker skin depth, and so might not require as small a diameter. Such possibilities are analyzed in detail in[18] with an emphasis on aluminum because of its lower cost compared to copper. In the case of a constraint on strand diameter, it is shown that the optimized performance is the same, using more strands of the higher-resistivity material. Thus, this approach is only attractive because it allows the use of a lower-cost material than copper—it does not allow improved performance, unless the material choice facilitates the wire-drawing manufacturing process for smaller diameter wire.

If instead, thin foil-like layers of copper are used, very thin layers can be used, with polymer laminates used for support if necessary[14]. The challenge then becomes developing strategies for interchanging the layer positions; some strategies for doing this have been developed[12,13,14]. A four-layer winding, for example, could provide up to a factor-of-two reduction in winding loss, based on the $1/\sqrt{p}$ factor from[15,16] discussed above. Although a factor-of-two improvement might sound like a small gain for the complexity required to make a four-layer winding, the analysis of air-core inductors in the following section shows that a factor-of-two improvement in loss can be translated into an order-of magnitude reduction in area with loss held constant.

At least in the short term, many practical designs in the 10 MHz to 100 MHz range will use single-layer copper windings, with resistance that is inversely proportional to the square root of frequency, because of skin effect. In an air-core inductor, the increase in inductive

impedance, proportional to frequency, outweighs the degradation that results from the increase in resistance, such that the quality factor $Q = \omega L/R$ increases with frequency[7,8]. Thus, despite the limitations of available winding options for addressing loss effects in this frequency range, increased frequency can still be attractive.

Air-Core Inductors

As explained in the previous section and in[7,8], the performance of air-core inductors improves as frequency is increased, and thus well designed air-core components can always do better as switching frequencies are increased. This is in contrast to magnetic-core components, which consistently have a maximum-performance frequency beyond which performance degrades, as will be discussed in the following two sections. The designer can choose whether to use the benefit of improved air-core performance at higher frequencies to improve efficiency, to decrease component size and cost, or both.

One approach to making an air-core inductor is to use a planar spiral coil, fabricated by photolithography in an integrated-circuit, printed-circuit-board, or other similar technology. This produces a field largely perpendicular to the substrate. From the perspective of eddy-current losses in the conductor, this is not attractive, because fields perpendicular to planar conductors induce much larger eddy currents than fields parallel to them. Moreover, the fields extend well above and below, out of the plane of the coil. The full volume occupied by this field must contain only high-resistivity materials to avoid the eddy-current losses that would be induced in any conductor. Even the choice of substrate requires special care to ensure that it is adequately resistive such that substrate eddy currents don't cause excessive losses[19,20]. EMI (electromagnetic interference) can also be a problem. If a conductive shield is used to control the field, current induced in the shield will create additional losses, although in some cases the losses in the shield may be smaller than the loss that would be induced in a higher-resistivity conductor nearby.

A solution to the external field problem is to use a toroidal air-core inductor[21,22], which can almost completely eliminate external fields. However, given that the field is only internal, the internal volume of the toroid must be substantial in order to create sufficient energy storage. Assuming a single-layer winding, both resistance and inductance scale approximately with the square of the number of turns, resulting in a quality factor that is roughly independent of the number of turns or of the specific inductance value targeted[21]. Thus, the most important design question is how the quality factor changes with the overall geometry. In[7] we showed that in the limit of using a large substrate area, but with limited height, a simple analysis of sheet resistance and its inductive equivalent, "inductance per square," $L_\square$, the limiting value of Q can be calculated as

$$Q_{max} = \frac{\omega L_\square}{2R_{\square,ac}} = \frac{\omega \mu_0 h}{2\rho/\delta}. \tag{3}$$

One way to summarize the results of that calculation is that for 1 mm thickness, one can obtain up to about 50 units of Q per millimeter thickness at 10 MHz; or 150 units per millimeter at 100 MHz. Although in an integrated circuit a 1 mm thickness may be too thick to be attractive, in many applications it would be considered nearly planar, and for many applications Q values on the order of 100 are adequate. However, (3) is the limit for a very large area. It does not address the question of how much area is required.

A simple analysis of the resistance and inductance of a toroid, without examining non-ideal factors in as much detail as is considered in[21], can provide additional insight on this design issue.

Toroid Analysis and Optimization

In our simplified design approach, we consider only a single-turn toroid, expecting to later choose an actual number of turns to scale up the inductance to the value needed, while approximately maintaining the same Q. Thus, we need only a simple calculation of the inductance and resistance of a single-turn toroid. The inductance, which is the same as the inductance of a coaxial line of length equal to the thickness of the substrate, is,

$$L = \frac{\mu_0 h}{2\pi} \ln(r_o/r_i), \tag{4}$$

where h is the thickness of the substrate, and r_o and r_i are the outer and inner radii.

The resistance is calculated by summing four segments, the top and bottom conductors (which have equal resistance), and the inner and outer cylinders. These resistances are, respectively,

$$R_t = R_b = \frac{\rho}{2\pi\delta} \ln(r_o/r_i), \tag{5}$$

$$R_i = \frac{hR_\square}{2\pi r_i} \tag{6}$$

$$R_o = \frac{hR_\square}{2\pi r_o} \tag{7}$$

where $R_\square = \rho/\delta$ is the is the sheet resistance (" resistance per square") for current flowing in the layer one skin depth thick.

Numerically evaluating Q for a given area for a range of different inner radii shows that Q is always maximized at $r_o/r_i \approx 0.2785$. With this ratio fixed, the only design choice for a given height is r_o, or equivalently, the area. Thus, we can plot Q for a complete set of design choices for a given substrate thickness and frequency as a function of area, as shown in Fig. 1. It is interesting to note that the inductance for the single-turn design does not change with the radius or area, but is fixed at $L = 0.2035\mu_0 h$ by the fixed ratio of inner and outer radii. The resistances R_t and R_b are also fixed, and the improvement in Q occurs purely through the reduction in R_o and R_i. The value of Q asymptotically approaches the limit (3) as R_o and R_i becomes small compared to R_t and R_b.

Solenoid Analysis and Optimization

An alternative to a toroid is a solenoid, also constructed with the main flux in the plane of the substrate, as shown in Fig. 2. This design does not fulfill our objective of keeping the flux entirely in the plane of the substrate, because when it comes out the ends, it returns to the other end through paths above and below the substrate. Nonetheless, it has advantages over a spiral coil in that the flux is somewhat contained and is largely parallel to the planar conductors, and, compared to a toroid, it has more closely coupled turns, and as a result it can achieve higher inductance for the same resistance. This advantage of the solenoid is well established for low-frequency, wire-wound air-core inductors[23,24,25]. We wish to examine quantitatively how much of an advantage in performance the solenoid offers in

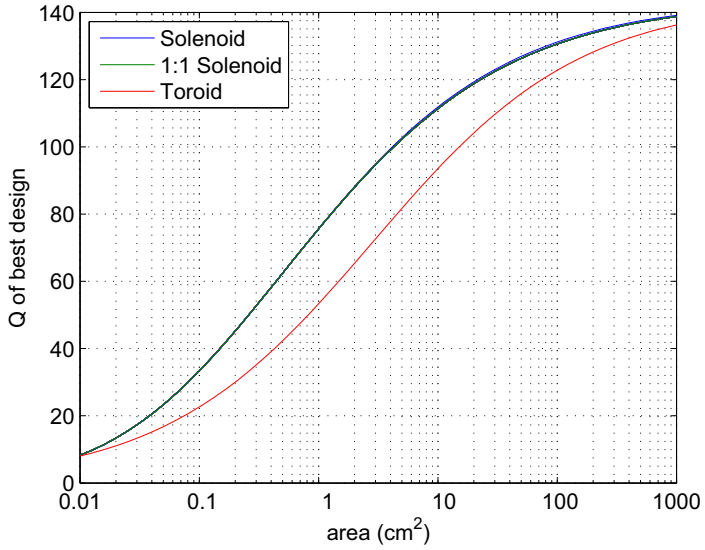


Figure 1: Best achievable quality factor Q for toroidal and solenoidal inductors as a function of area, for a 3-mm thick substrate with room-temperature copper conductors, at 10 MHz. See the text for scaling the results for different frequencies and thicknesses.

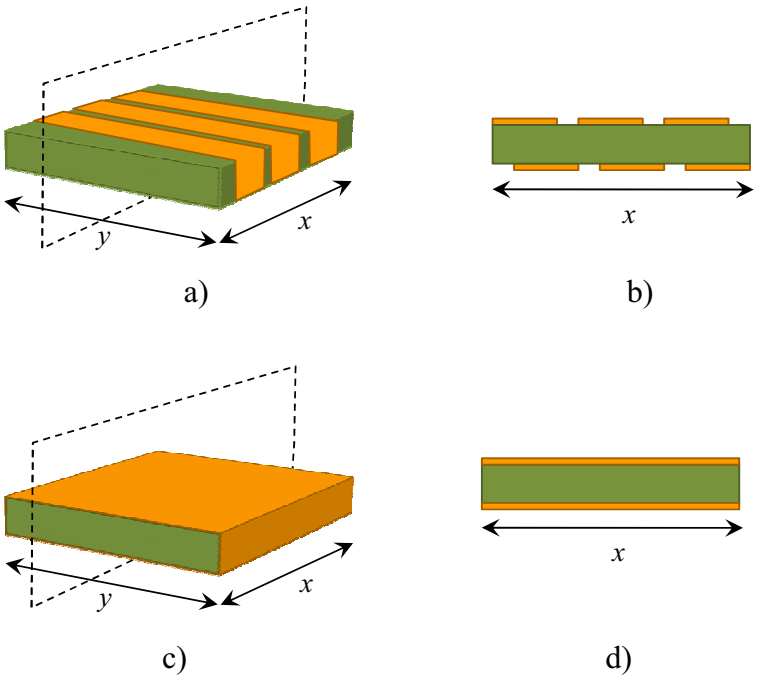


Figure 2: Solenoid geometry considered. a) Sketch of actual geometry. b) Cross-section, as indicated by the dashed outline in a). c) Single-turn configuration analyzed, for later scaling multiple turns. d) Cross-section of the single turn model.

high-frequency, height-constrained applications, in order to evaluate whether it is worth considering, even though it has a higher external field.

As with the toroid, we consider the model for a single-turn design, knowing that the

impedance can be scaled by the number of turns squared without a major effect on quality factor. A two-dimensional (2-D) cross-section through the single-turn design (Fig. 2 c) is shown in Fig. 2 d, taken as indicated by the plane outlined by a dashed line in Fig. 2 c.

Because of the lack of symmetry, accurately modeling a solenoid like that shown in Fig. 2 is more difficult than modeling a toroid. It is easy to model using approximations that are valid when the lateral dimensions, x and y, are large compared to the height, h, but as with a toroid, we wish to minimize the area used, and the degradation in performance that happens as the area is reduced is partly related to the deviation of the performance from what would be calculated from that simple analysis. Thus, in developing the model, it is important to consider what aspects of performance degrade as the lateral dimensions are reduced.

The simplest model for the single-turn design would be to calculate the inductance per square[7] and multiply by the ratio of lateral dimensions y/x to obtain $L = \mu_0 hy/x$ and use the resistance per square to obtain $R = 2R_{\square}y/x$. This model does not capture any of the effects that degrade performance as the lateral dimensions are reduced. According to the simple model, we could shrink x and y arbitrarily small, while maintaining the ratio y/x constant, and maintain constant inductance and resistance.

One factor that the simple model neglects is the resistance of the vertical sidewalls or "vias," visible on the right of Figs. 2a and 2c. When y is very large compared to the substrate thickness h, this resistance is negligible. However, as y gets smaller, it becomes significant. We can correct for this effect simply by adding the height to the length of the conductor in the resistance calculation:

$$R = 2R_{\square}(x + h)/y \tag{8}$$

The inductance also deviates from the simple model when the lateral dimensions get small, in particular, when x gets small compared to h. One way to derive the inductance per square is to approximate the magnetic reluctance of the path through the solenoid as $\mathcal{R} = \frac{x}{yh\mu_0}$ and assume that the reluctance of the return path through the air is negligible, because of the much larger area available for that path. However, this only holds when $x \gg h$. For smaller x, the inductance is less than would be predicted by the simple model because the return-path reluctance is no longer negligible, and calculating only based on the reluctance through the body of the solenoid is not adequate. Thus, we need a more accurate model for cases in which x is similar to or only moderately bigger than h.

We note that the configuration in Fig. 2 c) is similar to some types of transmission lines, and so we can use models from that literature. It is also a standard parallel-plate capacitor geometry, and, with no dielectric or magnetic materials, we can find the inductance per unit length of transmission line (y, into the page in Fig. 2 d) from the capacitance per unit length, based on the propagation speed in the transmission line being the speed of light, as

$$L' = \mu_0 \varepsilon_0 / C'. \tag{9}$$

Applying this to established equations for capacitance of this configuration[26] results in

$$L = \frac{2\mu_0 y}{2W_e/h + 1.393 + 0.667 \ln(2W_e/h + 1.444)} \tag{10}$$

for $(2W_e/h) \geq 1$, and

$$L = \frac{\mu_0}{\pi} \ln\left(\frac{4h}{W_e} + \frac{W_e}{2h}\right) \tag{11}$$

for $(2W_e/h) < 1$, where effective width W_e is the effective value of our x dimension, based on correcting for the conductor thickness t, calculated for two cases as

$$W_e = x + 0.398t\,(1 + \ln(h/t)) \tag{12}$$

for $x/h > \frac{1}{4\pi}$, and

$$W_e = \frac{1}{2\pi}\left[x + 0.398t\left(1 + \ln\left(\frac{4\pi x}{t}\right)\right)\right] \tag{13}$$

for $x/h \leq \frac{1}{4\pi}$. Spot checks comparing this to 2-D finite-element analysis indicated that it is highly accurate—certainly accurate enough to navigate the overall possible design space to find the correct region in which to operate.

With small x, the current distribution also becomes nonuniform, and the accuracy of (8) for predicting resistance is degraded. However, we found that the deviations from the simple model of inductance became significant at much larger values of x, such that modeling the effects of small x on inductance was the most important step to produce a valid design model, and that (8) provides an adequately accurate model of resistance.

Just as with the toroid, once the height and area are determined, there is one more variable available to optimize a single-turn inductor. In the toroid, that variable is the inner radius, and a constant ratio of inner radius to outer radius is optimal for any area. For the solenoid, the ratio x/y is the degree of freedom available. We numerically searched for the maximum value of Q, varying x/y while holding area constant. The resulting maximum Q for each area value examined is plotted in Fig. 1 along with the same results for toroids. Over this range of area constraints, the optimum aspect ratio x/y ranges between 0.71 and 2.4. However, for moderate areas, the optimum ratio is close to 1. Furthermore, for both large areas and small areas, Q varies very slowly with aspect ratio between one and the optimum value. Thus, using an aspect ratio of 1 for the full range of areas results in performance almost indistinguishable from the performance using the optimum value for each area. Both of these possibilities are included in Fig. 1, where they are indistinguishable for small areas, and barely distinguishable for areas greater than 10 cm^2.

Note that the inductance calculation (10) is based only on considering the two-dimensional cross section shown in Fig. 2 d. As the distance into the page in Fig. 2 d, y, becomes small, there is an opportunity for flux to curve out of the plane shown, as well as within the plane shown. This expanded flux path is effectively additional permeance in parallel with the return path that has been modeled, and will increase the inductance over what is calculated by the model.

Comparison of Toroid and Solenoid

From looking at Fig. 1, one can see that, although the toroid is always worse than the solenoid, the performance is never much worse. To make this comparison more explicit, the ratio of their maximum Q values is shown in Fig. 3, and one can see that the Q of the best toroid is never less than two thirds of the Q of the best solenoid. Given that the toroid essentially eliminates concerns about losses or EMI from the external field, this might seem a small price to pay. However, another way to compare the two is to look at the area requirement for the same Q value. Fig. 4 shows this comparison: starting with a toroidal design, how much smaller could one make a solenoid, and achieve the same performance? The answer is that, except for very small areas, one can achieve a reduction in area by better

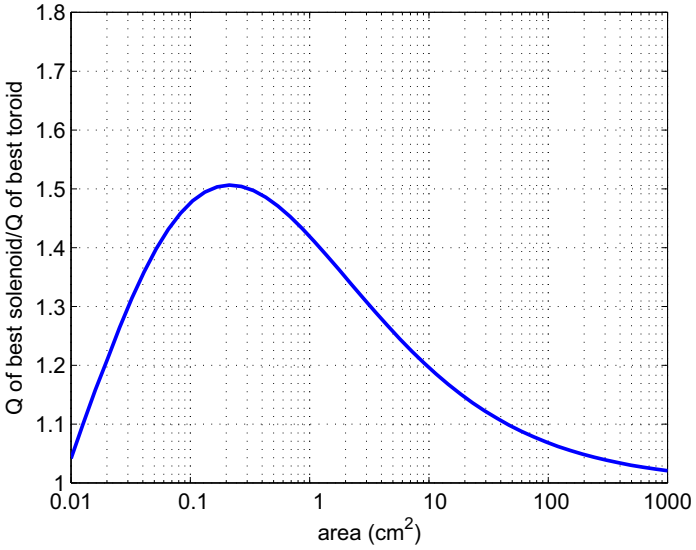


Figure 3: Ratio of the best achievable quality factor Q for solenoidal inductors to that for toroidal inductors as a function of area, for a three-millimeter thick substrate. The results can be applied to different substrate thicknesses by scaling the area axis by the square of the scaling factor for the substrate thickness.

than a factor of three. The different conclusion that one might draw from looking at the data in terms of area is a result of the very slow change in Q over a very large change in area: Fig. 1 shows only about one order of magnitude change in Q over five orders of magnitude change in area. Thus, we can conclude that the solenoid is well worth considering if its moderate external field can be tolerated.

Note that the accuracy of these comparisons is the lowest for very small areas: the information in the decade from 0.01 to 0.1 cm^2 should only be taken as a general indication of trends.

The data in these plots have been calculated based on 3 mm substrate thickness, copper conductor at room temperature, and a frequency of 10 MHz. Although it is not difficult to generate similar curves for other frequencies and substrate thicknesses, especially if one uses a fixed ratio of r_i/r_o = 0.2785 for the toroid, and an aspect ratio $x/y = 1$ for the solenoid, one can also scale the data in these curves to other frequencies and substrate thicknesses using the scaling formulas developed in[7],

$$Q \propto s\sqrt{f}, \tag{14}$$

where s is a scaling factor in linear dimensions. For example, suppose we want to consider a 1 mm thick substrate at a frequency of 40 MHz, and we want to achieve a Q value of 50 with a toroidal inductor. The dimensional scaling factor s is 1/3, and $\sqrt{f}$ goes up by a factor of two. Thus, we can potentially achieve a Q value of 2/3 the value shown on the plot. To achieve $Q = 50$, we need to locate a point for $Q = 75$ on the toroid curve in Fig. 1. That occurs at 3.4 cm^2. We need to scale the area by the square of the scaling factor $s^2 = 1/9$, to get an area of 0.38 cm^2, corresponding to an outer diameter of 7 mm.

A full design process would then choose a number of turns to realize the necessary inductance, adjust the size as needed to achieve the correct inductance with an integer

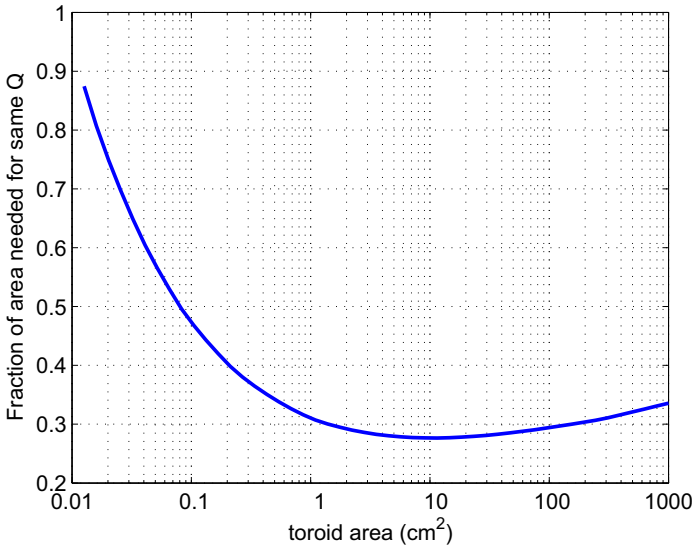


Figure 4: Ratio of the area required to achieve a given quality factor with a solenoidal inductor to the area required to achieve the same quality factor with a toroidal inductor. The horizontal axis is the area of the toroidal inductor for a three-millimeter thick substrate. The results can be applied to different substrate thicknesses by scaling the area axis by the square of the scaling factor for the substrate thickness.

number of turns, and examine second-order effects on quality factor such as capacitance and the loss of conductive material when the single-turn conductor is divided into multiple turns, as considered in[21].

Ferrites for MHz Frequencies

An important result of the analysis in the previous section is that there are well defined limits for the minimum size of an air-core magnetic component for a given quality-factor requirement. Adding a lossless magnetic material could, in principle, improve Q by a factor equal to the relative permeability of the magnetic material. As magnetic materials with permeability in the thousands are readily available, this approach seems highly attractive. However, Q is also degraded by losses that occur in any real magnetic material, losses which invariably increase with frequency. At frequencies of tens to hundreds of kilohertz MnZn ferrites are one of the very few magnetic materials that have sufficiently low losses to be usable in power applications. However, beyond about 1 MHz, their losses increase too rapidly, canceling out any advantage of operating at a higher frequency. Some MnZn ferrites have attractive performance up to a few megahertz, but beyond that one must turn to NiZn ferrites or powdered metal materials. NiZn is generally considered inferior for power applications because of its lower saturation flux density. However, at megahertz frequencies, MnZn ferrites must be operated at extremely low AC flux densities to limit power loss to acceptable levels, making saturation less of a concern. The key factor determining power-handling capability is the power loss with large-signal excitation. Unfortunately, data on power loss above 5 MHz with substantial flux density is rarely available on material manufacturers' data sheets, and is not easy to measure.

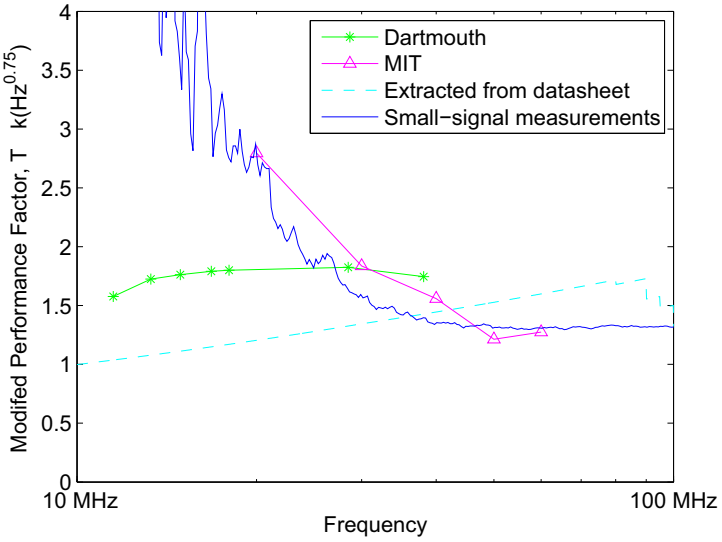


Figure 5: Modified performance factor $F_{mod} = \hat{B}(f) \cdot f^{0.75}$ for Ceramic Magnetics' N40 NiZn ferrite at power loss density of 500 kW/m^3. The "MIT" curve is based on [1,2] and the "Dartmouth" curve is based on our new measurements.

A measurement method and a set of measurement results for frequencies from 20 to 80 MHz are available in [1,2]. One particular material included in these measurements, N40 NiZn ferrite from Ceramic Magnetics, exhibited impressively low power loss at the low end of this frequency range. Motivated by this result, we undertook to further measure and analyze its performance.

A common and useful way to evaluate the frequency performance of power magnetic materials is through the use of performance factor, defined as $\hat{B}(f) \cdot f$ where $\hat{B}(f)$ is the amplitude of AC flux density that can be carried at a specified power loss density at the frequency f. If it is assumed that the current density that can be carried by a winding is fixed independent of frequency, then the power that can be handled by a transformer or inductor with a fixed amount of power loss is proportional to performance factor. A plot of performance factor can then show how the performance improves with increasing frequency and where it reaches its maximum. Beyond that point, performance factor decreases with increasing frequency, and there is no benefit to operating in this range.

As discussed in the section on winding losses, in many cases the resistance of a good winding design is proportional to the square root of frequency. A modified performance factor, $F_{mod} = \hat{B}(f) \cdot f^{0.75}$, accounts for this effect, and provides a more accurate assessment of the frequency dependence of power handling in components using a given material in the megahertz frequency range [27]. For this reason, we use modified performance factor to evaluate the potential of N40 NiZn ferrite.

Modified performance factor for N40 NiZn ferrite is shown in Fig. 5, with four different curves based on different data sources. Based on the data in [1,2], the highest measured performance factor is at 20 MHz, but the trend indicates that it would be much higher at lower frequencies, and no data for lower frequencies was included in [1,2]. We collected a data data set, using the method in [1,2], for the frequency range from 11 MHz to 38 MHz. Although the data for 30 MHz and above agrees well for both data sets, there is a significant

discrepancy near 20 MHz. Near 20 MHz, the new data indicates much lower performance, and it predicts nearly constant power handling capability across the frequency range from 15 MHz to 30 MHz. This trend is distinctly different from the steep slope in the plot based on the data it in [1,2].

Based on the observation that the loss in both measurements sets was approximately proportional to the square of the magnetic field strength, we conclude that the material is behaving approximately linearly in this range. If it had truly linear behavior, large-signal power-loss measurements would not be needed, and power performance could be predicted from small-signal measurements. For this reason, it was of interest to include predictions based on small-signal measurements in Fig. 5. We include a curve based on the N40 material datasheet, and a curve based on small-signal measurements with an Agilent E4991A impedance analyzer. Because this impedance analyzer has the capability for an extra phase calibration using a reference air capacitor to enhance the accuracy of high-Q measurements, we expect these measurements to be more accurate than the datasheet values. The performance predicted from these measurements is strikingly similar to that predicted from the measurements in [1,2], and predicts even higher performance factor at frequencies below 20 MHz. This might indicate that our new large-signal measurements are not accurate, and confirm the original measurements indicating very high performance factor at 20 MHz. However, we also note that the data in [1,2] indicates less linearity at 20 MHz than at higher frequencies. Thus, even if it is valid to extrapolate to large-signal performance from small-signal measurements at 30 MHz and above, it may not be valid in the 10 MHz to 20 MHz range. Thus, we can only conclude that further measurements in this range are needed.

Thin-Film Power Materials

Although NiZn ferrites have potential for high power density at tens of megahertz, even the most optimistic interpretation of the data in Fig. 5 indicates that practical operation is limited to 20 MHz or below at ac flux densities near 10 mT or below. For higher frequencies and higher flux densities, and thus higher power density, thin-film magnetic materials are more promising. They also offer more potential for integration. In this section, we briefly review the discussion in [7] of the potential these materials offer, and report recent results confirming their ability to perform well at high flux densities.

A salient difficulty with many thin-film metal-alloy magnetic materials is their low resistivity, leading to high eddy-current losses. Although using multiple laminated layers can mitigate this problem, it introduces fabrication complexity, and only works well if the flux lines stay strictly parallel to the layers. An attractive alternative is to use a material comprising nano-particles of metallic magnetic material separated by dielectric, similar in concept to conventional powdered iron materials, but with nm-scale particles [28 – 35]. The dielectric leads to high overall resistivity, while the metal grains are too small for significant eddy currents to circulate within them. Thin films of these materials can have anisotropic behavior that leads to low hysteresis losses. The losses can be low up to 100 MHz or more (Fig. 6) with higher flux density and temperature capability than ferrites.

Although nanogranular materials have much higher resistivity than metal-alloy thin film materials, their resistivity is still much lower than ferrites, and, when they are used in thicknesses of several μm or more, it can be advantageous to use a multi-layer structure to further reduce eddy currents. Although this is a standard technique, accurately modeling it

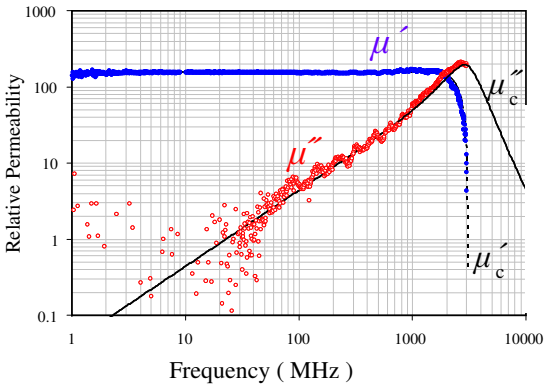


Figure 6: Complex relative permeability of a nano-granular Co-Zr-O magnetic film[35].

for very high frequencies requires including displacement currents that are not included in standard models[27,36].

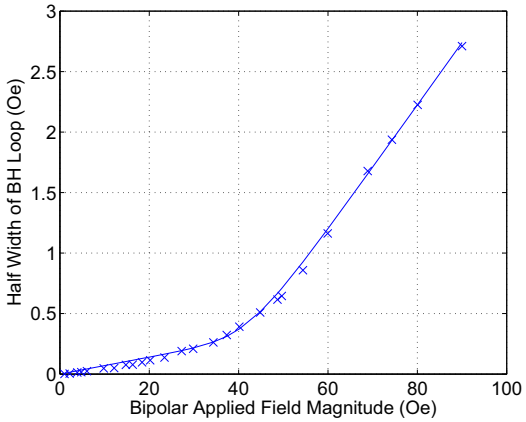


Figure 7: Hysteresis-loop half-width for a 19.3-μm-thick Co-Zr-O nanogranular magntic film measured at varying excitation levels in the hard-axis direction at 10 Hz.

Most nano-granular materials are deposited by a reactive sputtering thin-film deposition process, which typically produces films a few microns thick. Controlling deposition conditions and using a fine multilayer structure can facilitate consistently and robustly achieving good performance[34,37]. We have previously produced good quality films as thick as 10 μm[38], and applied these films in fabricating several types of inductors[5,6,39]. We have now extended this capability to films as thick as 19.3μm. As with thinner films, we achieved strong anisotropy by applying a magnetic field during deposition, resulting in low coercivity and narrow hysteresis loops in the "hard-axis" direction. As shown in Fig. 7, these loops are very narrow for excitation below about 40 Oe, corresponding to flux densities of about 0.35 T.

Although the combination of the large-signal B–H loop measurements like that shown in Fig. 7, performed at low frequency, and small-signal permeability measurements like that shown in Fig. 6 at high frequencies together indicates that high-frequency high-power performance will be good, a direct measurement under the conditions of a typical power application is preferable. We previously developed an instrument capable of measuring small

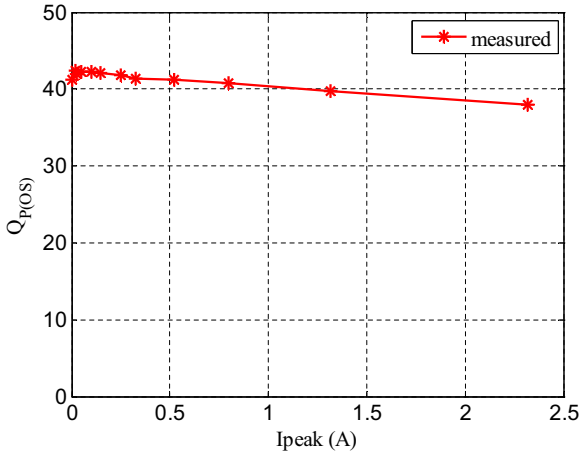


Figure 8: Quality factor of the inductors fabricated in[6] using nanogranular Co-Zr-O cores at 30 MHz as a function of current drive level, showing that the low-loss characteristics are maintained up to high drive levels, corresponding to about 0.14 T.

losses in samples of magnetic films with high-field 50 MHz excitation[40]. Although the results from measuring nanogranular films with that instrument indicated a small amount of nonlinear behavior that might be attributed to hysteresis increasing at high frequency, they also confirmed that total loss remains very low even with high-field high-frequency excitation. To add to this evidence, we have now been able to measure the inductors fabricated in[6] under large-signal high-frequency excitation using an approach similar to that described in[1,2]. Fig. 8 shows the resulting Q value of the component measured from the primary-winding terminals with the secondary open-circuited, as a function of current drive level. Very little variation occurs over the full range tested, which is up to a peak flux density of about 0.14 T.

Conclusion

In order to realize the benefits of increased switching frequency, enabled by wide-bandgap semiconductors, low-loss magnetics designs for frequencies in the megahertz range are needed. By 10 MHz, MnZn magnetic materials no longer perform well, and sufficiently small litz-wire strands become prohibitively expensive. We have examined a range of possible technologies, including single-layer windings, multi-layer windings, air-core inductors, NiZn ferrites, and nano-granular magnetic materials. All of these technologies show promise, although all of them stand to benefit from additional research.

Acknowledgements

This work was supported in part by grants from the United States National Science Foundation and the United States Department of Energy.

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